A Fixed Confidence Gate for Selective Classification

Anonymous submission

***Abstract*—Selective classification allows a model to abstain from uncertain predictions, but accuracy among accepted predictions describes a selected population. This manuscript examines a fixed confidence gate in a synthetic evaluation fixture, with coverage reported alongside conditional accuracy. The gate accepts predictions at or above a specified confidence threshold and otherwise abstains; the comparator uses the same classifier and accepts every prediction. Neither method retrains the model. Across two synthetic datasets, conditional accuracy increases on one and decreases on the other. These observations do not establish an overall accuracy improvement because the evaluation populations differ. A range bound places selective risk between its stated limits under bounded loss and positive coverage; it establishes neither a generalization guarantee nor superiority over the comparator. The fixture illustrates the need to interpret abstention metrics with their coverage and assumptions. Its values are not research findings. Statistical uncertainty, runtime, distribution shift, and deployment outcomes remain unevaluated, and missing classifier and data-generation details limit reproducibility.**

***Impact Statement*—Abstention changes which predictions contribute to a reported accuracy. Researchers and practitioners therefore need coverage to interpret accuracy among accepted predictions. This synthetic fixture illustrates that reporting only conditional accuracy can obscure the population being evaluated. It provides an example for checking whether manuscript language preserves this distinction during editing and template migration. The fixture is not evidence of a deployable classification system or an improvement in decision quality. No runtime, distribution-shift, or deployment assessment has been performed, and statistical uncertainty is unavailable. Any proposed practical use would require reproducible model and data descriptions, suitable comparisons, and evaluation of the relevant operating conditions. The numerical values remain synthetic inputs to a skill evaluation, rather than research findings or demonstrated social benefits.**

***Index Terms*—Confidence threshold, coverage, selective classification, selective risk**

I. Introduction

Selective classification allows a model to abstain from uncertain predictions. We evaluate a fixed confidence gate using synthetic fixtures. Because the gate selects which predictions are evaluated, we report coverage alongside accuracy among accepted predictions.

II. Method

Gate uses a fixed classifier and accepts a prediction when its maximum class probability is at least 0.8; otherwise, it abstains. Always uses the same classifier and accepts every prediction. Neither method retrains the model. For a fixed classifier h and acceptance function $a(x) \in \{0, 1\}$, assume $0 \leq \ell(h(x), y) \leq 1$ and positive coverage $C = \mathbb{E}[a(X)] > 0$. Selective risk is

$$R_{\text{sel}} = \frac{\mathbb{E}[a(X)\ell(h(X), Y)]}{C}. \tag{1}$$

Theorem 1: Under these assumptions, $0 \leq R_{\text{sel}} \leq 1$.
This range bound provides neither a generalization guarantee nor a comparison between Gate and Always. Appendix A gives the full proof.

III. Synthetic Evaluation

Alpha and Beta each contain 1000 synthetic samples. Both methods use the same model and 5 evaluation seeds; results are seed means, with no variance estimates.

TABLE I
SYNTHETIC FIXTURE VALUES. ALWAYS ACCURACY USES ALL SAMPLES; GATE ACCURACY IS CONDITIONAL ON ACCEPTANCE. THESE VALUES ARE NOT RESEARCH FINDINGS.

Dataset	Always (%)	Gate (%)	Coverage (%)
Alpha	80.0	84.0	60.0
Beta	76.0	75.5	58.0

Table I shows a higher conditional accuracy on Alpha and a lower conditional accuracy on Beta. Different evaluation populations prevent interpreting these differences as an overall accuracy improvement. No significance tests, runtime measurements, or distribution-shift evaluations were performed.

IV. Conclusion and Future Work

The synthetic evaluation compares a fixed gate with a single baseline. It does not establish state-of-the-art performance or a guarantee of lower risk. The classifier and data-generation details remain missing and are required for reproducibility. Conditional accuracy and coverage describe the accepted population; deployment benefits remain untested. Future work should assess uncertainty, runtime, and distribution shift after the reproducibility gaps are resolved. The source draft used the ICML author guidelines [1]; this version uses the locally verified IEEE TAI template.

References

[1] "Icml 2026 author instructions," https://icml.cc/Conferences/2026/AuthorInstructions, 2026.

Appendix A
Proof of the Range Bound

Since $0 \leq a(X)\ell(h(X), Y) \leq a(X)$, taking expectations gives $0 \leq \mathbb{E}[a(X)\ell(h(X), Y)] \leq C$. Division by $C > 0$ proves Theorem 1. This argument applies to the specified distribution; it does not establish a bound under distribution shift.